

170. Reconciliation Sublayer (RS) and Media Independent Interface for 800 Gb/s (800GMII)

170.1 Overview

This clause defines the characteristics of the Reconciliation Sublayer (RS) and the 800 Gb/s Media Independent Interface (800GMII) between Ethernet media access controllers and various 800 Gb/s PHYs. Figure 170–1 shows the relationship of the RS and 800GMII to the ISO/IEC OSI reference model.

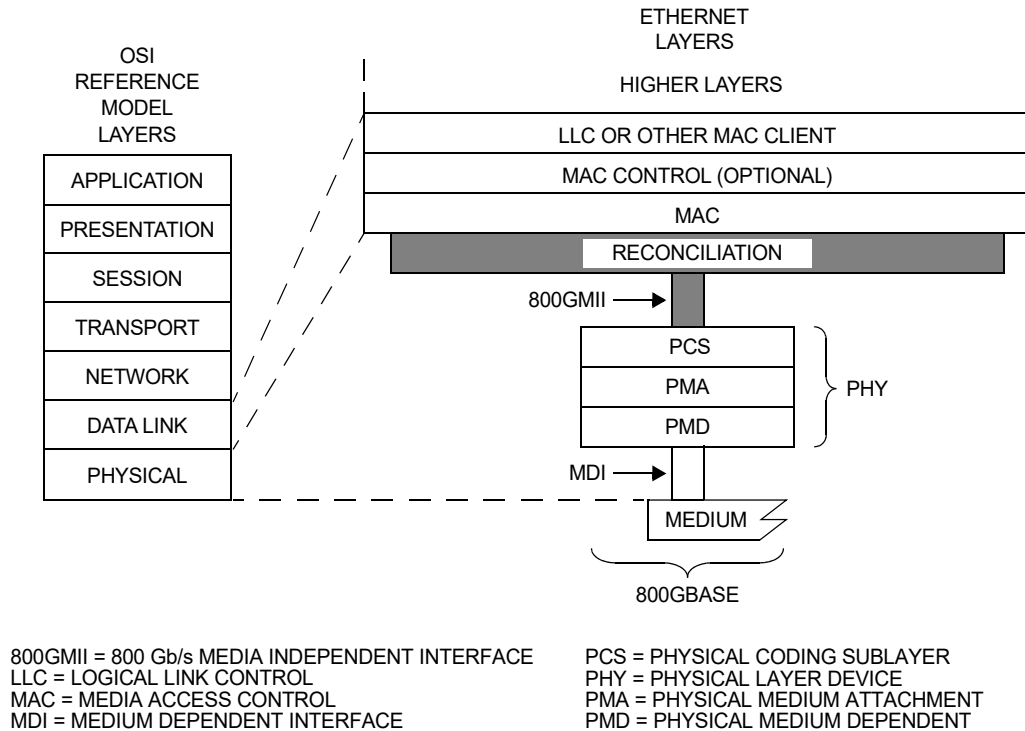


Figure 170–1—Relationship of the Reconciliation Sublayer and 800GMII to the ISO/IEC Open Systems Interconnection (OSI) reference model and IEEE 802.3 Ethernet model

The 800GMII is an optional logical interface between the MAC sublayer and the Physical Layer device (PHY). The 800GMII Extender may optionally be used to extend the 800GMII (see Clause 171).

The RS adapts the bit serial protocols of the MAC to the parallel format of the PCS service interface. Though the 800GMII is an optional interface, it is used in this standard as a basis for specification. The Physical Coding Sublayer (PCS) is specified to the 800GMII, so if not implemented, a conforming implementation shall behave functionally as if the RS and 800GMII were implemented.

The 800GMII has the following characteristics:

- a) It supports a speed of 800 Gb/s.
- b) Data and delimiters are synchronous to a clock reference.
- c) It provides independent 64-bit wide transmit and receive data paths.
- d) It supports full duplex operation only.

170.1.1 Summary of major concepts

The following are the major concepts of the 800GMII and RS:

- The 800GMII is functionally similar to other media independent interfaces that have been defined for lower speeds, as they all define an interface allowing independent development of MAC and PHY logic.
- The RS converts between the MAC serial data stream and the parallel data paths of the 800GMII.
- The RS maps the signal set provided at the 800GMII to the PLS service primitives provided at the MAC.
- Each direction of data transfer is independent and serviced by data, control, and clock signals.
- The RS generates continuous data or control characters on the transmit path and expects continuous data or control characters on the receive path.
- The RS participates in link fault detection and reporting by monitoring the receive path for status reports that indicate an unreliable link, and generating status reports on the transmit path to report detected link faults to the DTE on the remote end of the connecting link.

170.1.2 Application

This clause applies to the interface between the MAC and PHY, or between the PHY 800GXS and the PHY when the optional 800GMII Extender is used. This logical interface is used to provide media independence so that an identical media access controller may be used with supported PHY types.

170.1.3 Rate of operation

The 800GMII is specified to support 800 Gb/s operation.

170.1.4 Delay constraints

The maximum cumulative MAC Control, MAC, and RS delay (sum of transmit and receive delays at one end of the link) shall meet the values specified in Table 170–1. A description of overall system delay constraints and the definitions for bit times and pause_quanta can be found in 169.4 and its references.

Table 170–1—Delay constraints

Sublayer	Maximum (bit time)	Maximum (pause_quanta)	Maximum (ns)
800 Gb/s MAC, RS, and MAC Control	196 608	384	245.76

170.1.5 Allocation of functions

The allocation of functions at the 800GMII balances the need for media independence with interface simplicity. The 800GMII provides media independence by separating the Data Link and Physical Layers of the OSI seven-layer reference model.

170.1.6 800GMII structure

The 800GMII structure is identical to the CGMII structure specified in 81.1.6.

170.1.7 Mapping of 800GMII signals to PLS service primitives

The Reconciliation Sublayer (RS) shall map the signals provided at the 800GMII to the PLS service primitives defined in Clause 6. The PLS service primitives provided by the RS and described here behave in exactly the same manner as defined in Clause 6. Full duplex operation only is implemented at 800 Gb/s, therefore, PLS service primitives supporting CSMA/CD operation are not mapped through the RS to the 800GMII. This behavior and restrictions are the same as described in 22.2.1, with the details of the signaling described in 170.3.

The RS maps all primitives in an identical manner as the CGMII does and as specified in 81.1.7 except that EEE and LPI signaling are not supported.

170.2 800GMII data stream

The 800GMII data stream has the same characteristics as the CGMII data stream described in 81.2.

170.3 800GMII functional specifications

The 800GMII functions identically to the CGMII specified in 81.3 except that EEE and LPI signaling are not supported.

170.4 Protocol implementation conformance statement (PICS) proforma for Clause 170, Reconciliation Sublayer (RS) and Media Independent Interface for 800 Gb/s operation (800GMII)¹¹

170.4.1 Introduction

The supplier of a protocol implementation that is claimed to conform to Clause 170, Reconciliation Sublayer (RS) and Media Independent Interface for 800 Gb/s operation (800GMII), shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in Clause 21.

170.4.2 Identification

170.4.2.1 Implementation identification

Supplier ¹	
Contact point for inquiries about the PICS ¹	
Implementation Name(s) and Version(s) ^{1,3}	
Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s) ²	
NOTE 1—Required for all implementations. NOTE 2—May be completed as appropriate in meeting the requirements for the identification. NOTE 3—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).	

170.4.2.2 Protocol summary

Identification of protocol standard	IEEE Std 802.3df-2024, Clause 170, Reconciliation Sublayer (RS) and Media Independent Interface for 800 Gb/s operation (800GMII)
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No [] Yes [] (See Clause 21; the answer Yes means that the implementation does not conform to IEEE Std 802.3df-2024.)	

Date of Statement	
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¹¹Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

170.4.3 Major capabilities/options

Item	Feature	Subclause	Value/Comment	Status	Support
*MII	PHY support of 800GMII	170.2, 170.3		O	Yes [] No []

170.4.4 PICS proforma tables for Reconciliation Sublayer (RS) and Media Independent Interface for 800 Gb/s operation (800GMII)

170.4.4.1 General

Item	Feature	Subclause	Value/Comment	Status	Support
G1	Cumulative MAC Control, MAC and RS delay	170.1.4	Per Table 170–1	M	Yes []
G2	Lane structure	170.1.6	Per Table 81–2	M	Yes []

170.4.4.2 Mapping of PLS service primitives

Item	Feature	Subclause	Value/Comment	Status	Support
PL1	Mapping to Clause 6	170.1.7	RS implements mapping to Clause 6 PLS service primitives	MII:M	Yes [] N/A []
PL2	Mapping of PLS_DATA.requests	170.1.7	In sequence TXD<0> to TXD<63>	MII:M	Yes [] N/A []
PL3	Start control character creation	170.1.7	First octet of preamble converted to Start control character	MII:M	Yes [] N/A []
PL4	TXD and TXC generation	170.1.7	For each 64 PLS_DATA.requests	MII:M	Yes [] N/A []
PL5	Terminate control character creation	170.1.7	DATA_COMPLETE causes creation of Terminate control character in next lane in sequence	MII:M	Yes [] N/A []
PL6	Mapping RXD to PLS_DATA.indications	170.1.7	Create PLS_DATA.indications in sequence from RXD<0> to RXD<63>	MII:M	Yes [] N/A []
PL7	PLS_DATA.indication generation	170.1.7	Generate 64 PLS_DATA.indications for each RXD<63:0> until Terminate then generating 0, 8, 16, 24, 32, 40, 48, or 56	MII:M	Yes [] N/A []
PL8	Start control character conversion	170.1.7	Convert valid Start control character to preamble before generating PLS_DATA.indications	MII:M	Yes [] N/A []
PL9	Terminate control character	170.1.7	No PLS_DATA.indications generated	MII:M	Yes [] N/A []

Item	Feature	Subclause	Value/Comment	Status	Support
PL10	PLS_DATA_VALID.indication generation	170.1.7	On change of value of DATA_VALID_STATUS	MII:M	Yes [] N/A []
PL11	DATA_VALID_STATUS	170.1.7	Value of DATA_VALID on a lane 0 Start control character preceded by eight idles, a Sequence ordered set, or a Terminate character	MII:M	Yes [] N/A []
PL12	DATA_VALID_STATUS	170.1.7	Value of DATA_NOT_VALID on any control character but Error	MII:M	Yes [] N/A []
PL13	Frame not ending with Terminate control character	170.1.7	Ensure MAC detects CRC error	MII:M	Yes [] N/A []

170.4.4.3 Data stream structure

Item	Feature	Subclause	Value/Comment	Status	Support
DS1	Frame transfer	170.2	Within 800GMII data stream	MII:M	Yes [] N/A []
DS2	Bit mapping	170.2	Per Figure 81–4	MII:M	Yes [] N/A []
DS3	Content of <data>	170.2	Consist of data octets	MII:M	Yes [] N/A []
DS4	Recognition of <efd>	170.2	Terminate recognized in any lane	MII:M	Yes [] N/A []

170.4.4.4 800GMII signal functional specifications

Item	Feature	Subclause	Value/Comment	Status	Support
FS1	TX_CLK active edge	170.3	TXD and TXC sampled on the rising edge of TX_CLK	MII:M	Yes [] N/A []
FS2	TX_CLK frequency	170.3	One-sixty-fourth of the MAC transmit data rate	MII:M	Yes [] N/A []
FS3	TXC assertion and de-assertion	170.3	De-asserted for data, asserted for control character	MII:M	Yes [] N/A []
FS4	TXC clock	170.3	Synchronous to TX_CLK	MII:M	Yes [] N/A []
FS5	TXD encoding	170.3	Per Table 81–3	MII:M	Yes [] N/A []
FS6	TXD clock	170.3	Synchronous to TX_CLK	MII:M	Yes [] N/A []
FS7	Start alignment	170.3	Start control character aligned to lane 0	MII:M	Yes [] N/A []
FS8	RX_CLK active edge	170.3	RXD and RXC sampled on the rising edge of RX_CLK	MII:M	Yes [] N/A []

Item	Feature	Subclause	Value/Comment	Status	Support
FS9	RX_CLK frequency	170.3	One-sixty-fourth of the MAC receive data rate	MII:M	Yes [] N/A []
FS10	Loss of receive signal	170.3	Source RX_CLK from nominal clock	MII:M	Yes [] N/A []
FS11	RXC assertion and de-assertion	170.3	De-asserted for data, asserted for control character	MII:M	Yes [] N/A []
FS12	RXC clock	170.3	Synchronous to RX_CLK	MII:M	Yes [] N/A []
FS13	RXD decoding	170.3	Per Table 81–4	MII:M	Yes [] N/A []
FS14	RXD clock	170.3	Synchronous to RX_CLK	MII:M	Yes [] N/A []
FS15	Received Error control character	170.3	RS cause MAC FrameCheckError	MII:M	Yes [] N/A []
FS16	DATA_VALID assertion	170.3	RS not assert DATA_VALID unless Start control character in lane 0	MII:M	Yes [] N/A []

170.4.4.5 Link fault signaling state diagram

Item	Feature	Subclause	Value/Comment	Status	Support
LF1	Link fault signaling state diagram	170.3	Implement per Figure 81–11	MII:M	Yes [] N/A []
LF2	link_fault = OK and MAC frames	170.3	RS services MAC frame transmission requests	MII:M	Yes [] N/A []
LF3	link_fault = OK and no MAC frames	170.3	In absence of MAC frames, RS transmits Idle control characters	MII:M	Yes [] N/A []
LF4	link_fault = Local Fault	170.3	RS transmits continuous Remote Fault Sequence ordered sets	MII:M	Yes [] N/A []
LF5	link_fault = Remote Fault	170.3	RS transmits continuous Idle control characters	MII:M	Yes [] N/A []